8/11

38+2
40A

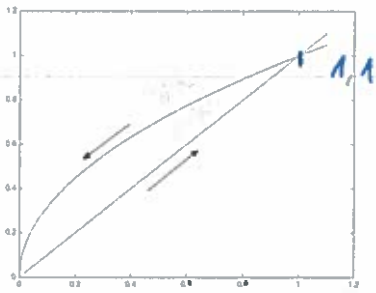
EE2-08C Vector Calculus CLASS TEST

Please answer each question by clearly circling your answer, i.e. circling (a), (b), (c), (d) or (e). Note that some of the questions do not have a choice (e). Each correct answer will earn 4 marks. Each incorrect answer will receive -1 marks. A question left blank will receive 0 marks.

PLEASE WRITE YOUR NAME AND STUDENT NUMBER IN THE BOX BELOW:

THOMAS COTTIN 01 06 1787

1. The figure below shows a closed contour C, given by the line $y = x$ from the origin to the point $(1, 1)$, and the curve $x = y^2$ from $(1, 1)$ back to the origin.



$\partial x = 6y^2 - 3x^2$

The value of the line integral over this contour: $\oint_C (5x^2 + 2y^3)\,dx + (6xy^2 - x^3)\,dy$ is

(a) $\frac{3}{20}$, (b) $-\frac{3}{14}$, (c) $-\frac{3}{28}$, (d) $-\frac{3}{10}$,

(e) none of the above.

∂y

$6y^2$

2. Consider the function $\phi(x, y, z) = \frac{x^2 y}{z^3} + xe^{-z}$ at the point $P(1, 0, -1)$. What is the value of $\nabla\phi$ at P?

(a) $e + 2$, (c) $\frac{1}{e}\boldsymbol{i} + \boldsymbol{j} - e\boldsymbol{k}$, (b) $\frac{2}{e} - 1$, (d) $e\boldsymbol{i} - \boldsymbol{j} + e\boldsymbol{k}$,

(e) None of the above

3. Consider the vector field $\boldsymbol{F} = (y^2 - \sin x)\boldsymbol{i} + xy^2\boldsymbol{j} + (5 - z^3)\boldsymbol{k}$, and

$$I = \oint_C \boldsymbol{F} \cdot d\boldsymbol{r}\,,$$

where C is the unit circle: $x^2 + y^2 = 1$. Then the value of I is given by

(a) $\frac{\pi}{2}$, (b) 0, (c) $-\pi$, (d) $-\frac{\pi}{2}$

(e) None of the above

4. Define $\boldsymbol{G} = (2y^2 - \sin x)\boldsymbol{i} + 4xy\boldsymbol{j} + (1-2z)\boldsymbol{k}$, and $\boldsymbol{G} = \nabla\phi$, with $\phi(0,0,0) = 0$. Then the functional form of ϕ is given by:

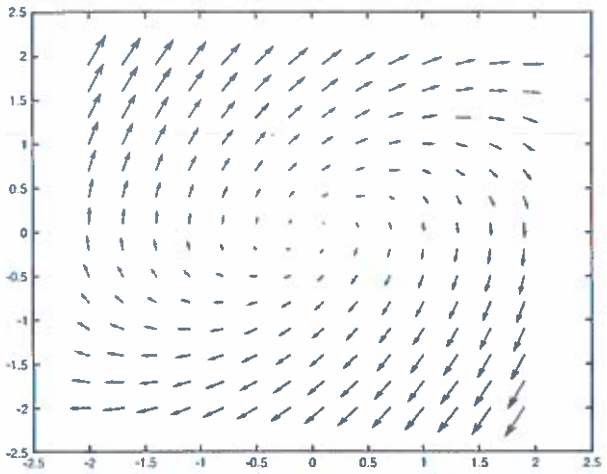
(a) $2xy^2 + \cos x - z(1-z)$, (b) $(2xy^2 + \cos x)\boldsymbol{i} + 2xy^2\boldsymbol{j} + z(1-z)\boldsymbol{k}$,

(c) $2xy^2 - \cos x + z(1-z)$, (d) ϕ does not exist, as $\boldsymbol{G}$ is curl-free,

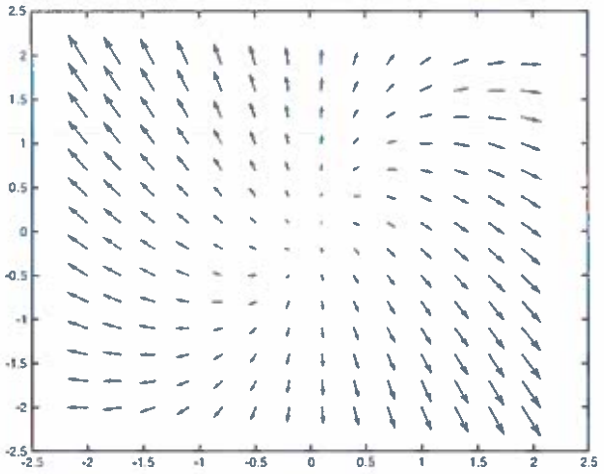
(e) None of the above

5. Which of the following figures shows the 2-D field

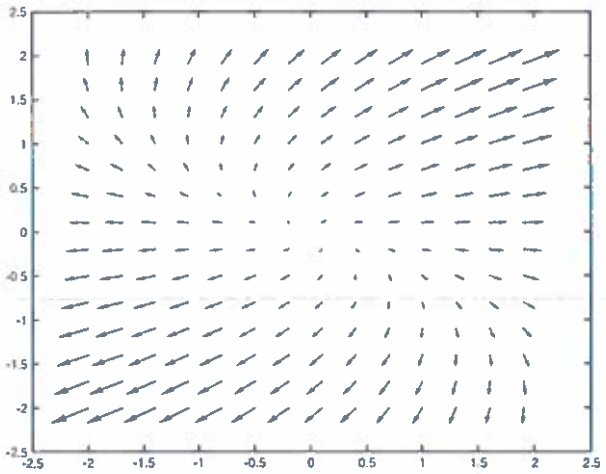
$$\boldsymbol{F} = x\boldsymbol{i} + (x+y)\boldsymbol{j}\,?$$



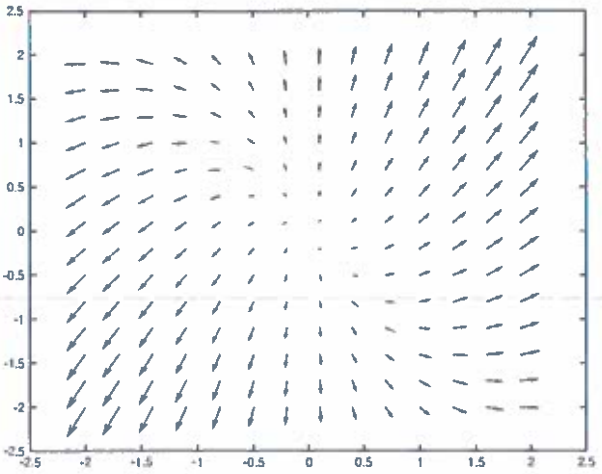
(a)



(b)



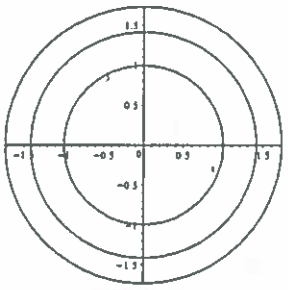
(c)



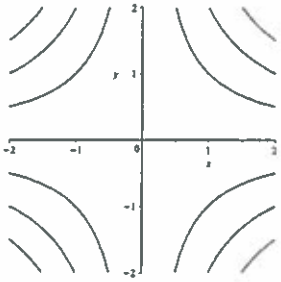
(d)

6. Which of the following figures shows the field lines for

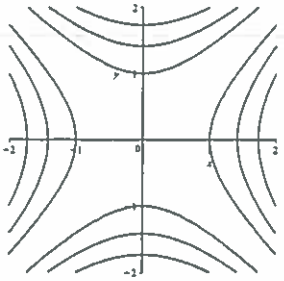
$$\boldsymbol{F} = y\boldsymbol{i} - x\boldsymbol{j}\,?$$



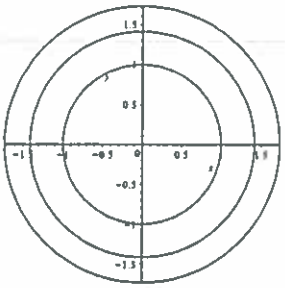
(a) (clockwise)



(b)



(c)



(d) (anti-clockwise)

7. The volume $\mathcal{V}$ is the cube with one corner at the origin and another at the point $(-1, 1, 1)$. The (closed) surface of $\mathcal{V}$ is denoted by S. Define $\boldsymbol{G} = x^3y^2z^2\boldsymbol{i} - 2x^2y^3z^2\boldsymbol{j} + x^2y^2z^3\boldsymbol{k}$ and

$$I = \iint_S \boldsymbol{G} \cdot \mathrm{d}\boldsymbol{S}\,.$$

Then the value of I is given by:

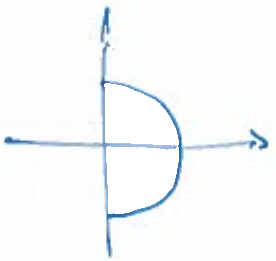
(a) $2\boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$, (b) 2, (c) 4π , (d) 0 ,

(e) None of the above

8. Consider the line integral $I = \int_A^B ds$, along the anticlockwise curve $x^2 + y^2 = 1$, from $A = (0, -1)$ to $B = (0, 1)$. The value of I is given by:

(a) 0, (b) π, (c) $-\pi$ (d) 2π,

(e) None of the above



9. Consider the curve $\boldsymbol{r} = t\boldsymbol{i} + t^3\boldsymbol{j} + t^2\boldsymbol{k}, \quad (0 \le t \le 1)$. Define the function

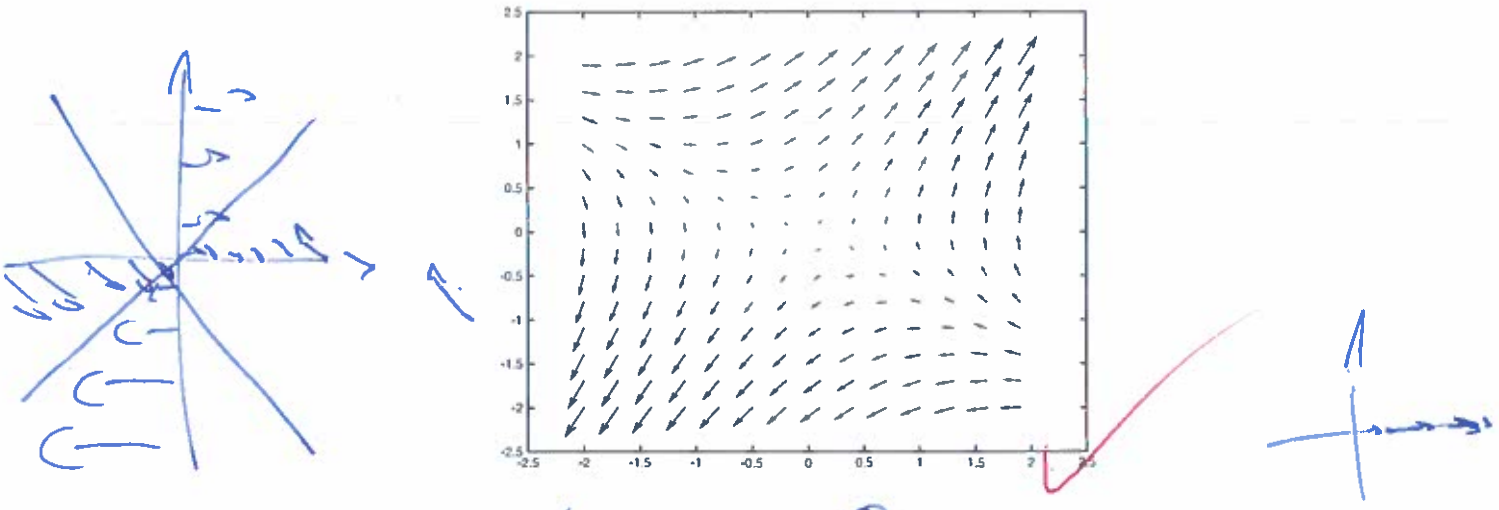
$$\boldsymbol{F} = (y - 2xz)\boldsymbol{i} + 2x\boldsymbol{j} + (z - 2x^2)\boldsymbol{k}\,,$$

and the integral $I = \int_A^B \boldsymbol{F}\cdot d\boldsymbol{r}$, where A corresponds to $t = 0$ and B corresponds to $t = 1$. The value of the integral I is:

(a) $\frac{1}{2}\boldsymbol{i} + \boldsymbol{j} - \frac{3}{4}\boldsymbol{k}$, (b) $\frac{1}{4}$, (c) $2\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$, (d) $\frac{3}{4}$,

(e) None of the above

10. Which of the following 2-D fields is shown illustrated in the figure below?



(a) $\boldsymbol{F} = (y - x)\boldsymbol{i} + x\boldsymbol{j}$, (b) $\boldsymbol{F} = x\boldsymbol{i} + (x - y)\boldsymbol{j}$, (c) $\boldsymbol{F} = y\boldsymbol{i} + (x + y)\boldsymbol{j}$, (d) $\boldsymbol{F} = (x + y)\boldsymbol{i} + y\boldsymbol{j}$,

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Green's Theorem in a plane. Let $P(x, y)$ and $Q(x, y)$ be arbitrary, continuous functions in the $x - y$ plane, in which there is a closed, piecewise smooth boundary C enclosing a region R. Then

$$\oint_C (P\,dx + Q\,dy) = \int\int_R (Q_x - P_y)\,dx\,dy\,.$$

Divergence (Gauss') Theorem Given an arbitrary 3D vector field $\boldsymbol{u}(x, y, z)$ in a finite, simply connected volume V with closed surface is S, where dS is an element of area on S, then

$$\int\int\int_V \operatorname{div} \boldsymbol{u}\, dV = \iint_S \boldsymbol{u}\cdot \mathrm{d}\boldsymbol{S} = \iint_S \boldsymbol{u}\cdot \hat{\mathbf{n}}\, dS$$

where $\hat{\mathbf{n}}$ is a unit vector normal to the surface S.

Stokes' Theorem Given an arbitrary 3D vector field $\boldsymbol{v}(x, y, z)$ and a surface S with boundary C,

$$\oint_C \boldsymbol{v}\cdot d\boldsymbol{r} = \int\int_S \operatorname{curl} \boldsymbol{v}\cdot \hat{\mathbf{n}}\, dS = \int\int_S (\nabla \times \boldsymbol{v})\cdot \mathrm{d}\boldsymbol{S}$$

where $\hat{\mathbf{n}}$ is a unit vector normal to the surface S.

Formulae

$$
\begin{aligned}
\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} &= a_x b_x + a_y b_y + a_z b_z \,, \\
\underline{\mathbf{a}} \times \underline{\mathbf{b}} &= (a_y b_z - a_z b_y)\underline{\mathbf{i}} + (a_z b_x - a_x b_z)\underline{\mathbf{j}} + (a_x b_y - a_y b_x)\underline{\mathbf{k}} \\
\nabla &= \frac{\partial}{\partial x}\underline{\mathbf{i}} + \frac{\partial}{\partial y}\underline{\mathbf{j}} + \frac{\partial}{\partial z}\underline{\mathbf{k}} \,. \\
\nabla\phi &= \frac{\partial \phi}{\partial x}\underline{\mathbf{i}} + \frac{\partial \phi}{\partial y}\underline{\mathbf{j}} + \frac{\partial \phi}{\partial z}\underline{\mathbf{k}} \\
\nabla \cdot \underline{\mathbf{F}} &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \,, \\
\nabla \times \underline{\mathbf{F}} &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\underline{\mathbf{i}} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\underline{\mathbf{j}} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\underline{\mathbf{k}} \\
\nabla^2\phi &= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \,. \\
\nabla \cdot (\phi \underline{\mathbf{F}}) &= \underline{\mathbf{F}} \cdot \nabla\phi + \phi \nabla \cdot \underline{\mathbf{F}} \\
\nabla \cdot (\phi \nabla\psi - \psi \nabla\phi) &= \phi \nabla^2\psi - \psi \nabla^2\phi \,, \\
\nabla \times \nabla \times \underline{\mathbf{F}} &= \nabla(\nabla \cdot \underline{\mathbf{F}}) - \nabla^2 \underline{\mathbf{F}}
\end{aligned}
$$